

Zaid Hasan

zaidhasan@gmail.com | (713) 530-1534 | [linkedin.com/in/zaidhasan2278](https://www.linkedin.com/in/zaidhasan2278) | github.com/zaid2278 | zaidhasan.vercel.app | Emerson, NJ

EDUCATION

BS Computer Science – New Jersey Institute of Technology, Newark, NJ

Sep 2022 - May 2026

GPA – 3.37 | Awards: NJIT Academic Merit Scholarship; Dean's List: Fall '22, Spring '23, Fall '23, Spring '24, Spring '25, Fall '25, Spring '26

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, C++, JavaScript/TypeScript, SQL, HTML/CSS, PHP, React.js, Node.js, NestJS, Adobe ColdFusion, Vite

Cloud & Infrastructure: AWS (EC2, ECS), Azure, Oracle Cloud (OCI), Docker, Linux, Git, Jira, PostgreSQL, MongoDB, Redis, Tableau, Power BI

Architecture & Security: REST APIs, Microservices, WebSockets (Socket.io), JWT, bcrypt, Swagger, Microsoft Presidio, Cloudflare, ETL (Power Query), ERD

AI/ML & Tools: Llama 3.1, vLLM, OpenAI API, Scikit-learn, Pandas, NumPy, AI Coding Assistants (Cursor, Copilot, Claude, ChatGPT), ServiceNow

WORK EXPERIENCE

Software Developer Intern | DentTracks

Jun 2026 – Present

- Architect and develop DentCommLayer, a real-time staff communication system built on a NestJS/TypeScript REST API, Redis pub/sub for cross-service message routing, WebSockets (Socket.io) for live notifications, and MongoDB, forming a distributed, multi-tier design built for high availability.
- Orchestrate an autonomous AI agent (AI Ninja) using Python 3.11, Llama 3.1 70B/8B models, and vLLM on AWS EC2 GPU instances to resolve complex dental billing issues, applying machine learning to real-world data mining and information retrieval problems and reducing manual review workload.
- Containerize the application stack with Docker for scalable AWS ECS deployment and implement data quality and governance procedures, enforcing HIPAA compliance via Microsoft Presidio to scrub Protected Health Information (PHI) prior to LLM processing.

Software Engineer Intern | RDE Systems

Jan 2026 - May 2026

- Engineered a secure, full-stack Medication Management Web Application using React, an Adobe ColdFusion REST API, and Microsoft Azure SQL, translating healthcare stakeholder requirements into a cloud-native data storage and processing architecture.
- Implemented strict role-based access control using JWT authentication and bcrypt to securely isolate patient portals, provider dashboards, and digital prescription systems, ensuring data accuracy and reliability across environments.
- Authored ER diagrams and data-flow diagrams alongside the live cloud architecture to document system design decisions, supporting enterprise-grade reliability and streamlining future migration planning for the engineering team.

IST Service Desk Student Assistant | NJIT Digital Learning and Campus Support

Jan 2024 - May 2026

- Resolved trouble tickets in ServiceNow and applied advanced troubleshooting skills to address complex technical issues for 13,000+ students and staff, collaborating cross-functionally with full-time IT staff for prompt, high-satisfaction outcomes.

Software Engineer Intern | USA Dental Solutions

Jun 2023 - Sep 2023

- Contributed to the DentTracks SaaS platform by developing APIs to aggregate patient data into centralized databases, supporting seamless synchronization, data warehousing needs, and HIPAA compliance.
- Engineered secure file processing workflows for automated reporting and integrated Tableau to visualize real-time KPIs on intuitive analytics dashboards for stakeholders.

PROJECTS

CareerOS

PERN Stack: React.js, Node.js, PostgreSQL, OpenAI API, Cloudflare, Sentry

- Designed UI/UX prototypes and developed the Job Statistics dashboard to visualize application data for users on a full PERN (PostgreSQL, Express/Node, React) stack.
- Optimized application performance by implementing lazy loading and code splitting to maximize Lighthouse scores.
- Hardened security using Cloudflare caching and integrated Sentry for real-time uptime monitoring and error tracking in production.

Marine Navigation System

Microsoft Maps API, Marine Traffic API, PHP, HTML, CSS, SQL, JavaScript

- Crafted a dynamic, reliable web application for real-time ocean mapping, processing and displaying hundreds of global ship location data points from live APIs.
- Architected an automated ETL pipeline using PHP to extract raw maritime telemetry from the Marine Traffic API, transforming and loading structured spatial data into an SQL database for real-time querying.

CERTIFICATIONS

Power BI and Tableau for Data Visualization

- Power BI: Mastered end-to-end analytics using Power Query for ETL processes (merging, pivoting) and designed complex data models (Star and Snowflake schemas) using DAX functions.
- Tableau: Specialized in advanced visualization techniques, creating interactive dashboards with calculated fields and custom hierarchies to translate raw data into strategic business insights.

Oracle Cloud Infrastructure 2025 AI Foundations Associate

- Gained practical insight into deploying large language models, orchestrating data workflows, and managing scalable AI infrastructure within cloud environments.